

7(a)(i)	$(92 - 35) \div 35 \times 100$ OR $(57 \div 35) \times 100$; 163 (%) ;	2 ECF answer to 3 s.f. from correct working with mis-read graph figures (from bars or trend line) e.g. if $(63 - 3) \div 3 \times 100$ ALLOW 2000 ECF
7(a)(ii)	LIST RULE APPLIES but IGNORE incompletely named factors any two pairs of a factor (F) and reason (R) from: 1F <u>light intensity</u> ; 2R limits / affects, rate of, light-dependent stage / photophosphorylation / photoactivation ; 3F carbon dioxide <u>concentration</u> ; 4R limits / affects, rate of, light independent stage / Calvin cycle / carboxylation / carbon fixation ; 5F <u>temperature</u> ; 6R limits / affects, rate of, (named) enzyme-catalysed reactions / formation of enzyme-substrate complexes / Calvin cycle / light independent stage / photolysis ; 7F AVP ; 8R AVP ;	4 ALLOW 1 mark for 'limiting factor for / limits <u>rate</u> of / affects <u>rate</u> of, <u>photosynthesis</u>' if 2R / 4R / 6R not given 2R ALLOW limiting factor for 'limits rate of' 2R ALLOW description e.g. if light intensity is higher photophosphorylation, increases / occurs more / is faster 4R ALLOW limiting factor for 'limits rate of' 4R ALLOW description 6R ALLOW limiting factor for 'limits rate of' 6R ALLOW description 6R IGNORE high temperature denatures enzymes 6R CON low temperature denatures enzymes 7F (concentration in water of) nitrate / phosphate / (other named) mineral ion / mineral nutrients / dissolved oxygen OR fertiliser run-off / sewage discharge (into water / ocean) OR (water / ocean) pH 8R linked to, role in metabolism / named organic molecule

Question	Answer	Marks	Guidance
7(b)	any four from: 1 <u>sodium ions / Na⁺</u>, do not enter, <u>motor</u> neurone / <u>axon</u> ; 2 no, propagation / transmission / passing on, of, depolarisation / action potential / impulse ; 3 no Ca²⁺ enter, synaptic knob / presynaptic neurone ; 4 no acetylcholine, release / exocytosis ; 5 no, stimulation of / depolarisation of / binding of acetylcholine to, sarcolemma ; 6 no, <u>T-tubule</u>, depolarisation / action potential / impulse transmission ;	4	IGNORE SR, troponin, tropomyosin, actin, myosin ECF after first error of each type: 'into membrane', wrong ion symbol, omitting to state ion ECF for 'less' instead of 'no' after first error 1 ALLOW Na⁺ moves in at ligand-gated channels only 1 DO NOT ALLOW Na⁺ enter membrane for 'enter neurone' 1 DO NOT ALLOW toxin, mimics ACh / binds to ACh receptors (i.e. affects ligand-gated channel) 2 ALLOW impulse, not sent to / doesn't reach, synaptic knob 3 ALLOW Ca²⁺ channels don't open at, synaptic knob / presynaptic neurone 4 ALLOW ACh 4 ALLOW acetylcholine does not diffuse across, cleft / neuromuscular junction / synapse 5 ALLOW muscle cell membrane / postsynaptic membrane, for 'sarcolemma' 5 DO NOT ALLOW membrane of postsynaptic knob / postsynaptic neurone 5 DO NOT ALLOW toxin, mimics acetylcholine / binds to acetylcholine receptors (i.e. affects ligand-gated channel) but ECF from mp1